An Explainable Transfer Learning Framework for Early Detection of Diabetic Retinopathy Using Deep Learning

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Declaration

We hereby declare that this thesis/project report entitled **"An Explainable Transfer Learning Framework for Early Detection of Diabetic Retinopathy Using Deep Learning"** is our original work and has not been submitted before for any degree or diploma at this or any other university.

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It is hereby certified that the research work presented in this thesis entitled, "An Explainable Transfer Learning Framework for Early Detection of Diabetic Retinopathy Using Deep Learning," is the result of the original work carried out by Saif Hossain Likhon, Noman Ahmed, Mustafizur Rahman, and Dipto Nandi. Furthermore, it is certified that this thesis/report satisfies the requirements and standards for the degree of Bachelor of Science in Computer Science and Engineering.

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Dedication

We dedicate this research to our loving parents, whose unwavering love, encouragement, sacrifices, and constant support have been the foundation of our academic journey. Their guidance and belief in our abilities have inspired us to overcome challenges and pursue excellence. This work is also dedicated to our families, teachers, and well-wishers for their continuous motivation and encouragement throughout the completion of this thesis.

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Abstract

Diabetic retinopathy (DR) is a retinal disease linked to diabetes. It can cause serious vision loss. Early screening can help find the disease at an earlier stage. This work presents an explainable deep learning framework for five-class DR grading from retinal fundus images. The study uses the APTOS 2019 Blindness Detection dataset.The code uses a stratified 80/20 split for training and validation. Each image is read with OpenCV. It is changed from BGR to RGB and resized to 256×256 pixels. Gaussian blur based image enhancement is then used. Training images use horizontal flip, vertical flip, and random 90-degree rotation. Validation images use normalization without these random changes. WeightedRandomSampler is used to reduce the effect of class imbalance in the training set. Six ImageNet-pretrained CNN models are trained. They are EfficientNet-B0, ResNet34, MobileNetV3-Large, DenseNet121, VGG16-BN, and ConvNeXt Tiny. The final layer of each model is changed to five outputs. Training uses Cross-Entropy Loss, AdamW, a learning rate of 5×10^{-4}, weight decay of 10^{-4}, cosine annealing, batch size 32, and a maximum of 15 epochs. Early stopping uses a patience of four epochs.The final validation reports give 81% accuracy for EfficientNet-B0, 78% for ResNet34, 83% for MobileNetV3-Large, 78% for DenseNet121, 80% for VGG16-BN, and 81% for ConvNeXt Tiny. The code selects ConvNeXt Tiny and MobileNetV3-Large for the ensemble because they have the two highest observed validation accuracy values during training. Their softmax outputs are averaged with equal weight. The ensemble gives 83% accuracy, macro precision 0.70, macro recall 0.71, and macro F1-score 0.70. Grad-CAM is then applied to ConvNeXt Tiny to produce visual explanations. The framework therefore joins model comparison, transfer learning, class-balanced sampling, probability averaging, and explainable AI in one research pipeline.

Keywords: *Diabetic Retinopathy; Deep Learning; Transfer Learning; Convolutional Neural Network; Medical Image Classification.*

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List of Abbreviations and Acronyms

AI	Artificial Intelligence
CNN	Convolutional Neural Network
DR	Diabetic Retinopathy
FN	False Negative
FP	False Positive
Grad-CAM	Gradient-weighted Class Activation Mapping
TN	True Negative
TP	True Positive

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Chapter 1

1 Introduction

Diabetic retinopathy (DR) is a retinal complication linked to diabetes. It can damage blood vessels in the retina. It may produce signs such as microaneurysms, hemorrhages, exudates, and new blood vessels. Early DR may not show clear symptoms. Regular screening is therefore important.Manual grading of fundus images needs trained experts. Large screening programs may require many images to be checked. This can take time. It can also create differences between observers. Deep learning offers another way to analyze retinal images. CNNs can learn useful image features directly from images.However, automated DR grading still has several limits. Retinal images can differ in focus, light, color, and lesion appearance. The five DR stages are also not equally represented in the data. Nearby stages can look similar. This can lead to class confusion. Earlier studies also show that data choice, class balance, and model design can affect DR results [1].These limits motivate the present work. The study uses one common pipeline to compare six pretrained CNNs. It uses weighted sampling to reduce the effect of class imbalance. It also uses a two-model probability-average ensemble. Grad-CAM is added to inspect model focus. These steps do not remove all limits. They aim to reduce some practical limits within the data and code used in this study.

1.1 Problem Statement

The main problem is to classify retinal fundus images into five DR stages while handling class imbalance and image variation. A second problem is to compare several CNN designs under the same broad conditions. A third problem is to provide a visual explanation after prediction. The implementation must also fit within practical GPU memory and training time limits.

The notebook addresses these needs through a fixed image size, online augmenta-

tion, weighted random sampling, transfer learning, early stopping, and a probability-level ensemble. The framework is designed as a research system for computer-aided screening. It is not presented as a replacement for clinical diagnosis.

1.2 Problem Background

Earlier DR studies show that deep learning can learn useful signs from retinal images. These studies show that CNN-based models can support automated DR classification and screening. The reported results also show that model choice can affect the final classification result, as different CNN models can give different levels of accuracy [1, 2]. Transfer learning has also been used for DR image classification, which shows that pre-trained deep learning features can be useful for this task [3]. Other studies show that DR detection and DR stage grading can be different tasks. Stage grading can be harder when the signs of two nearby stages look similar [4]. Class imbalance can also affect DR grading because some DR classes may have fewer images than others [1]. These findings show that the problem is not only about choosing a deep learning model. It also includes class balance, image variation, stage overlap, model choice, and fair model testing. Clear model output is also needed so that the model result can be checked and understood. This study deals with these issues within the APTOS 2019 dataset. It uses a common training and test setup for six pre-trained CNN models. This allows the models to be compared under the same main conditions. The study also uses a probability-average ensemble to combine the outputs of the selected models.

1.3 Research Objectives

The primary objective of this research is to develop an automated and interpretable framework for early DR detection. The specific objectives are:

- To assess and compare the performance of six state-of-the-art pretrained convolutional neural networks (EfficientNet-B0, ResNet34, DenseNet121, MobileNetV3-Large, VGG16-BN, and ConvNeXt Tiny) for multiclass diabetic retinopathy classification.

- To develop an ensemble learning framework that combines the best models for improved prediction robustness and classification accuracy.
- Incorporating Gradient-weighted Class Activation Mapping (Grad-CAM) for visualisation of the regions of retina contributing to classification decisions, to improve the interpretability and clinical reliability of the model.

1.4 Motivation

The motivation comes from the need for fast and repeatable image screening. A trained model can process many images with the same code path. Transfer learning also gives a useful starting point because the six networks already contain visual features learned from ImageNet. The code then adapts their final layer to the five DR classes.The second motivation is model diversity. Different CNN designs may learn different image patterns. The code therefore does not rely on one architecture. It first compares six models. It then combines the two models that show the highest observed validation accuracy during training.The third motivation is explainability. The final pipeline uses Grad-CAM on ConvNeXt Tiny. The heatmap gives a visual view of the features that affect a selected class score. It is used as an explanation aid. It is not treated as clinical proof.

1.5 Significance of the Research

This research is significant because it studies five-class DR grading with six CNN designs in one controlled implementation. It also records class-level metrics. This is important because overall accuracy can hide weak results on less common stages.The work also shows how class balancing can be built into a training pipeline. WeightedRandomSampler gives higher sampling probability to classes with fewer training examples. This allows the training loader to expose the models to a more balanced stream of samples.The ensemble and Grad-CAM add two further views. The ensemble combines two probability vectors. Grad-CAM provides a visual explanation from the selected model. Together, these steps give a wider analysis than a single accuracy value.

1.6 Research Contributions

The main contributions of this work are summarised as follows:

- A comprehensive comparison of six state-of-the-art transfer learning architectures for multiclass diabetic retinopathy classification.
- Development of an ensemble framework combining MobileNetV3-Large and ConvNeXt Tiny to improve prediction robustness.
- A WeightedRandomSampler-based class-balancing method for the training loader.
- Integration of Grad-CAM visualization for explainable diabetic retinopathy diagnosis.

1.7 Complex Engineering Analysis

The project involves several complex engineering problem parameters and complex engineering activities.

1.7.1 Complex Engineering Problems

The project aligns with the seven complex engineering problem parameters as follows:

- **Specialist knowledge:** The work needs knowledge of CNNs, transfer learning, image processing, class imbalance, optimization, ensemble learning, and explainable AI.
- **Conflicting requirements:** The design must balance classification quality, GPU memory, training time, image size, and model size.
- **High-level analysis:** The study needs analysis of class imbalance, model behavior, validation results, model selection, and Grad-CAM output.
- **Unfamiliar problems:** Five-class retinal grading has stage overlap and small classes. These create cases that are not simple image classification cases.

- **Interdependent factors:** Data partitioning, augmentation, sampling, model choice, loss, optimizer, stopping rule, and ensemble choice affect the final system together.
- **Novel use of engineering tools:** The code combines PyTorch, TorchVision, OpenCV, Albumentations, Pandas, NumPy, and scikit-learn in one research pipeline.
- **Consequences and limits:** A wrong class may affect a screening workflow. Thus, the system is kept as a research and decision-support tool, not an independent clinical decision maker.

1.7.2 Complex Engineering Activities

The project also aligns with the five complex engineering activity parameters:

- **Problem analysis:**We analyzed the challenges of five-class grading, class imbalance, image variation, stage overlap, and explainability needs.
- **Design and development:** A full pipeline is designed from CSV labels and image loading to six CNNs, model selection, ensemble prediction, and Grad-CAM.
- **Use of modern tools:** The work uses Python, PyTorch, TorchVision, OpenCV, Albumentations, Pandas, NumPy, and scikit-learn.
- **Testing and evaluation:** The models are tested on the same stratified validation set with accuracy, precision, recall, F1-score, class reports, and confusion matrices.
- **Engineering judgment and communication:** Results are compared with care. Limitations are recorded. Peak validation accuracy is kept separate from final checkpoint accuracy.

1.8 Thesis/Project Organization

The remainder of this document is organized as follows. Chapter 2 reviews studies related to diabetic retinopathy detection and the main CNN methods used in this work. Chapter 3 presents the research methodology, data split, image processing, model analysis, ensemble method, and Grad-CAM process. Chapter 4 presents the implementation and validation results. Chapter 5 discusses standards, constraints, ethics, social impact, and project challenges. Chapter 6 gives the conclusion, limitations, and future work.

Summary

This chapter introduced diabetic retinopathy and the need for automated retinal image screening. It described the main problems in manual grading and automated grading. It also explained image variation, class imbalance, stage overlap, and explainability limits. The objectives, motivation, significance, and contributions of the work were then presented. Finally, the seven complex engineering problems and five complex engineering activities were linked to the project.

Chapter 2

2 Background Study and Literature Review

Deep learning is now an important method for medical image analysis. CNNs can learn image patterns without a manual feature design stage. Transfer learning can reduce the training burden by starting from weights learned on a large image set. This study uses that idea with six pretrained CNNs.

2.1 Literature Review

Prior DR studies demonstrate that deep learning can learn useful signs from retinal images. CNN-based methods have been applied to detect and classify DR. The results also suggest that model choice, data type, class balance and task design can affect the final result.

Pratt et al. used a CNN with image augmentation for DR classification. Their work showed that CNNs can learn useful signs from fundus images. The study also showed that DR class gaps can make grading hard. The reported accuracy was 75% [1].

Masood et al. used pre-trained deep learning features for DR image classification. Their study reported an accuracy of 48.2%. This shows that pre-trained models can be used for DR tasks. However, the result can change with the model, data, and class setup [3].

Kwasigroch et al. used deep CNN models for both DR detection and stage assessment. Their study reported 82% accuracy for DR detection and 51% for stage assessment. These results show that DR detection and DR stage grading are not the same task. Stage grading can be hard when nearby stages have close retinal signs [4].

Wang et al. compared different CNN models for DR stage classification. They tested AlexNet, VGG16, and Inception-V3. The reported accuracy was 37.43% for

AlexNet, 50.03% for VGG16, and 63.23% for Inception-V3. These results show that the CNN design can affect multi-class DR grading [2].

Harun et al. used an artificial neural network for fundus image classification. Their study reported 72.11% training accuracy and 67.47% testing accuracy. The gap between the two results shows why a model should also be tested on data that is not used for training. This is important for a fair view of model performance [5].

Suriyal et al. used a deep neural network for mobile-assisted DR detection. Their study reported a final accuracy of 73.3%. This result shows that deep learning can support automated retinal image analysis. However, DR detection is not the same as five-class DR grading. The task and class setup can affect the result [6].

Qummar et al. studied ensemble learning for DR detection. They combined the outputs of several CNN models. Their method reported an accuracy of 80.8%, with a recall of 51.5% and an F1-score of 53.74%. This shows that more than one model can be used for final DR prediction [7].

Dondeti et al. used deep convolutional features for fundus image classification. Their study used NASNet features with a non-linear embedding step and reported an accuracy of 77.90%. This shows that deep features can support automated fundus image classification. It also shows that feature choice and the final classification step can affect the result [8].

Dekhil et al. used deep learning for computer-aided DR diagnosis. Their study reported an accuracy of 77%. This shows that deep learning can be used to support automated DR analysis. However, the result can vary with the model, data, and test setup used in the study [9].

Taufiqurrahman et al. used a hybrid deep learning method for DR classification. They combined MobileNetV2 with an SVM classifier. The reported accuracy was about 79.01%. This shows that a deep CNN can be combined with another classifier for DR grading. It also shows that the final classification method can affect model performance [10].

Bodapati et al. studied multi-modal deep CNN features for DR severity prediction. Their study used blended deep features and reported an accuracy of 80.96%.

This shows that features from more than one deep model can be useful for DR severity prediction. It also supports the use of combined model features for retinal image analysis [11].

Gangwar and Ravi used transfer learning and deep learning for DR detection on the APTOS 2019 dataset. Their study reported an accuracy of 82.18%. This result is close to the result of the proposed work. It also shows that transfer learning can be useful for DR analysis on the APTOS 2019 data [12].

Bodapati et al. used a composite deep neural network with a gated attention method for DR severity classification. Their study reported an accuracy of 82.54%. This shows that attention-based deep models can learn useful features for DR grading. However, such models use a different design from the standard pre-trained CNN models tested in this study [13].

Kumar et al. developed the DRISTI method for DR diagnosis. Their method reported a five-class validation accuracy of 75.50%. This result shows that hybrid models can be used for multi-class DR diagnosis. It also shows the need to test model performance across more than one class [14].

Li et al. developed the CANet model for joint DR and diabetic macular edema grading. The model reported a joint accuracy of 65.1% on the IDRiD dataset. This shows that one deep model can be used for more than one retinal disease task. However, the task and dataset are different from the five-class APTOS 2019 grading used in this study [15].

Yaqoob et al. combined deep features from ResNet with a random forest classifier for DR detection. Their study reported a five-class accuracy of 75.09% on the EyePACS dataset. This shows that deep features can also be used with a separate machine learning classifier. The result also shows the effect of dataset choice on DR classification [16].

Shaban et al. used a 2D CNN for automated DR staging. Their study reported a testing accuracy of 80.2%. This shows that CNNs can support multi-stage DR classification. It also shows the need for a separate test set when checking the final model result [17].

Youldash et al. studied deep learning for multi-class DR diagnosis. They used DenseNet-121 and reported an accuracy of 82.0%. This result shows that modern CNN models can support multi-class DR grading. It also provides a useful reference for comparing different pre-trained CNN models [18].

Jain et al. used VGG19 for DR detection and severity classification. Their reported five-class accuracy was 76.9%. This shows that a common CNN model can be used for multi-class DR grading. It also gives a reference for comparing different CNN architectures [19].

Hattiya et al. studied EfficientNet-B0 for DR stage classification. Their study reported a test accuracy of 81.30%. This shows that an efficient CNN model can support DR stage grading. It also provides a useful reference for the use of EfficientNet-based transfer learning in this study [20].

Overall, the reviewed studies show that deep learning can support DR detection, classification, and stage grading. Different studies use different datasets, class settings, CNN models, and test methods. The reported results also vary across these settings. Some studies use one CNN model, while others use transfer learning, hybrid models, attention methods, or model ensembles. These differences make direct comparison difficult.

A clear need remains for a common test setup where several pre-trained CNN models are compared on the same five-class dataset. Class balance also needs care because some DR classes have fewer samples. Model outputs should also be checked with a clear explanation method. This study addresses these points by using the APTOS 2019 dataset, weighted random sampling, six pre-trained CNN models, and a common training setup. It also uses a probability-average ensemble with MobileNetV3-Large and ConvNeXt-Tiny. Grad-CAM is used to inspect the image regions that affect the model output.

2.2 Problem Analysis

The literature shows useful results from deep learning, but the studies do not all use the same task, data, or test plan. Some focus on referable DR. Some focus

on lesion detection. Some use one main model. Others use different image sources or tasks. These differences make it hard to compare the results in a fair way.The present work addresses a clear and focused gap. It studies five-class DR grading with the APTOS 2019 dataset. It also gives a direct comparison of six pretrained CNN models. Class-balanced sampling is used to reduce the effect of class imbalance. A two-model ensemble is then used at the probability level. Grad-CAM is also used to show which image areas the model uses.The gap is not a claim that earlier systems are weak. The gap is about how the methods are used together. Earlier work often studies one model, one task, or one main method. The present work puts several useful methods into one clear code pipeline. The notebook also keeps each step easy to trace. It starts with data loading and ends with model output and final visual checks. To make the main problems clear, the key gaps found in the literature and the way this thesis addresses them are shown in Table 1.

Table 1: Identified gaps and how this thesis addresses them

Identified Gap	**How This Thesis Addresses It**
Different studies use different DR tasks and class settings. Some use binary detection, while others use multi-class grading.	This thesis uses a five-class DR grading task with the APTOS 2019 dataset. This provides one fixed task for comparing the models.
Class imbalance can affect model learning because some DR classes have fewer images than others.	This thesis uses weighted random sampling during training. This gives more weight to classes with fewer samples and helps reduce the effect of class imbalance.
Many studies focus on one main CNN model, while different CNNs can learn different image features.	This thesis compares six pretrained CNN models in the same training pipeline. It also uses a two-model probability-level ensemble to combine the outputs of selected models.
Errors can occur between close DR stages because some retinal signs can look similar.	This thesis compares the validation results of six CNN models and studies the class-level results. This helps show which parts of the five-class task remain difficult.
Model results alone do not show why a CNN makes a certain decision, and different studies may use different training and evaluation steps.	This thesis uses Grad-CAM to view the image areas linked to the model output and follows one clear training and validation pipeline for model comparison.

The table shows the main problems found from the reviewed work. The problems are linked to task choice, class balance, model choice, model combination, and model explanation. The present thesis brings these parts into one pipeline. This gives a

clear way to study five-class DR grading on APTOS 2019.The aim is not to replace all earlier methods. The aim is to test several useful ideas under one common setup. This makes the model study easier to follow. It also gives a clear link between the problems found in earlier work and the methods used in this thesis.

Summary

This chapter reviewed major studies related to diabetic retinopathy detection and grading. It covered CNNs, transfer learning, EfficientNet, ResNet, DenseNet, MobileNetV3, ConvNeXt, and Grad-CAM. It also reviewed work on data quality, class differences, and evaluation. The final section defined the research gap addressed by the present five-class APTOS pipeline.

Chapter 3

3 Research Methodology

Methodology is designed as a clear and structured process for five-class diabetic retinopathy grading. It describes the main steps from data preparation to model evaluation. The first things we load are the training files from the APTOS 2019 Blindness Detection competition. Labels are read from `train.csv`. Each PNG image is read by a custom PyTorch Dataset. The image is pre-processed before being fed to the model. These are then fed to six pretrained CNNs via training and validation loaders.

3.1 Overall System Framwork

This section describes the proposed framework for automatic detection and grading of diabetic retinopathy (DR) based on transfer learning and ensemble learning tech- niques. The overall workflow of the proposed framework is shown in Figure 3.1

The overall framework starts with the APTOS 2019 image data and the `train.csv` label file. First, the code maps each `id_code` to its PNG image. The diagnosis value is read as an integer from 0 to 4.

Second, the labeled data is split into training and validation parts with stratification. This keeps the class structure more consistent across the two parts.

Third, each image is read with OpenCV. It is converted from BGR to RGB. It is resized to 256×256 pixels. Gaussian blur based processing is then used for image enhancement.

Fourth, training images pass through random horizontal flip, random vertical flip, and random 90-degree rotation. Validation images do not use these random changes. Both sets use ImageNet-style normalization and tensor conversion.

Fifth, the training loader uses WeightedRandomSampler. The sampler gives more

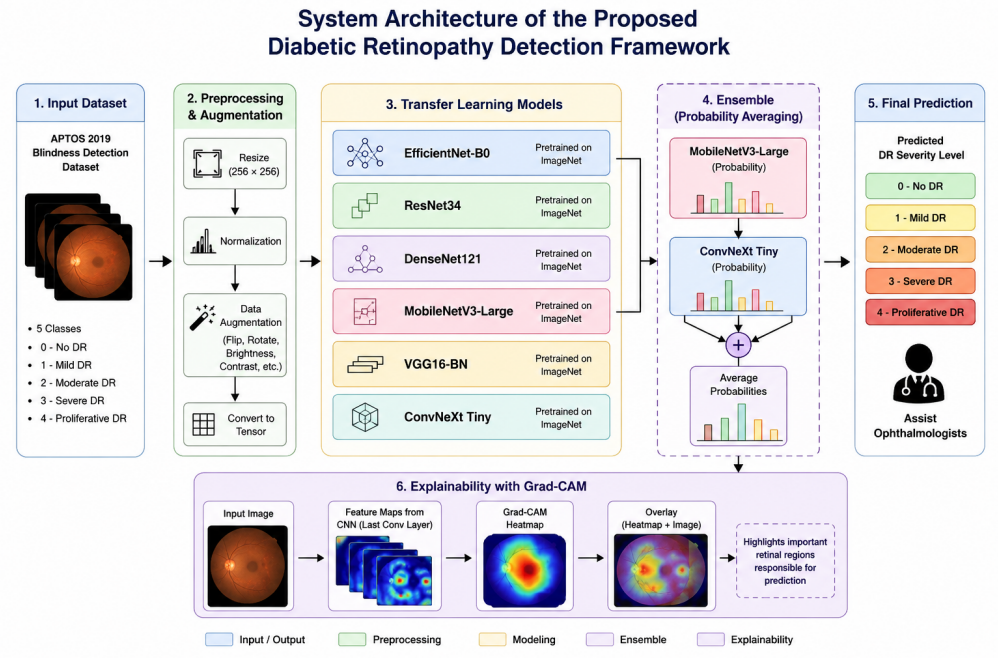


Figure 3.1: Overall architecture of the proposed diabetic retinopathy detection framework.

chance to classes with fewer training samples. This helps reduce the effect of class imbalance during training.

Sixth, six ImageNet-pretrained CNNs are trained. Each final classification layer is changed to five outputs. CrossEntropyLoss is used. AdamW uses a learning rate of 5×10^{-4} and weight decay of 10^{-4}. Cosine annealing is used for the learning rate. The batch size is 32. The maximum epoch count is 15. Early stopping uses four epochs of patience.

Seventh, the code records validation accuracy at each epoch. The highest observed validation accuracy is used to choose two models for the ensemble. ConvNeXt Tiny reaches 84.72%. MobileNetV3-Large reaches 83.49%. These two models are selected.

Eighth, each selected model produces five logits for each validation image. Softmax changes the logits into probability vectors. The two vectors are averaged with equal weight. The class with the largest mean probability becomes the final ensemble class.

Ninth, Grad-CAM is applied to ConvNeXt Tiny. A forward hook saves feature activations. A backward hook saves gradients. The code then creates a heatmap for

a selected class. The heatmap is resized and blended with the retinal image. One example from each of the five classes is used for visual output.

The main contribution is this linked pipeline. It does not use one model alone. It compares six pretrained CNNs. It handles class imbalance in the training loader. It selects two models from observed validation accuracy. It averages their softmax probabilities. It then adds Grad-CAM to inspect model focus.

3.2 Proposed Framework

The proposed framework has linked modules. The first module loads image and label data. The second module prepares each retinal image. The third module applies training-time augmentation and normalization. The fourth module trains six pretrained CNN models.The next module records validation accuracy during training. The two models with the highest observed validation accuracy are selected. In the current implementation, these are ConvNeXt Tiny and MobileNetV3-Large.The selected models then produce softmax probability vectors. The vectors are averaged with equal weight. The class with the largest mean probability is used as the ensemble output.The final module uses Grad-CAM on ConvNeXt Tiny. It creates a heatmap that shows regions that affect a selected class score. The heatmap is used for model inspection. It is not treated as proof of a clinical lesion.This design keeps model comparison, class balancing, ensemble prediction, and explanation in one pipeline. It also keeps final validation results separate from peak validation values used for ensemble selection.

3.3 Dataset and Label Structure

The code reads `train.csv` from the APTOS 2019 Blindness Detection directory. The image file name is formed from the `id_code` field by adding the `.png` extension. The diagnosis field is converted to an integer label.

The five labels are represented as Stage 0, Stage 1, Stage 2, Stage 3, and Stage 4. These are the five DR grades used in the APTOS task.

The validation report contains 733 images. The support is 361 for Stage 0, 74

Table 2: APTOS DR Class Labels

Stage	DR Class
0	No DR
1	Mild DR
2	Moderate DR
3	Severe DR
4	Proliferative DR

for Stage 1, 200 for Stage 2, 39 for Stage 3, and 59 for Stage 4. This distribution shows why class-aware training is needed.

3.4 Data Partitioning

The code uses `train_test_split` with `test_size=0.20`, `random_state=42`, and `stratify=df['diagnosis']`. Therefore, 80% of the available labeled data is used for training and 20% is used for validation.

Stratification is important because the five classes are not equally represented. It helps keep the class structure more consistent in both parts. The fixed random state also makes the split repeatable.

There is no separate test set in the notebook. The reported accuracy, precision, recall, F1-score, and confusion matrices are based on the validation set. These results are therefore described as validation results. They are not independent external test results.

3.5 Image Processing

The Dataset class first reads an image with OpenCV. The image is converted from BGR to RGB. It is then resized to 256×256 pixels.

After resizing, the code applies Gaussian blur with sigma value 10. It then combines the original image with the blurred image by using `cv2.addWeighted(image, 4, blurred, -4, 128)`. This creates an enhanced image before augmentation and normalization.

If OpenCV cannot read an image, the code uses a zero-valued $256 \times 256 \times 3$ array as a fallback. This keeps the data pipeline from stopping because of a read error. In

a clinical system, corrupted-image handling should instead be recorded and checked because replacing an image with zeros can affect input quality.

3.6 Training Augmentation and Normalization

Training images use three random geometric operations. HorizontalFlip is used with probability 0.5. VerticalFlip is used with probability 0.5. RandomRotate90 is used with probability 0.5.

The code then applies ImageNet-style normalization. The mean values are $(0.485, 0.456, 0.406)$. The standard deviations are $(0.229, 0.224, 0.225)$. ToTensorV2 converts the image into a PyTorch tensor.

Validation images do not use random flips or rotations. They use the same normalization values and tensor conversion. This keeps validation inference deterministic apart from the model itself.

3.7 Class Balancing

The code counts each diagnosis class in the training DataFrame. It calculates inverse class counts as class weights. Each training sample receives the weight of its class.

WeightedRandomSampler then samples training examples with replacement. The number of sampled examples is equal to the number of training samples.

This method does not change the validation set. It changes the chance of selecting training samples. Classes with fewer training images receive larger weights. This is a direct response to the uneven class support seen in the validation reports.

3.8 Algorithm and Model Analysis

The study uses transfer learning instead of training each CNN from scratch. Each model starts with ImageNet-pretrained weights. The original final classifier is then changed to five outputs.

Six models are evaluated. They are EfficientNet-B0, ResNet34, MobileNetV3-Large, DenseNet121, VGG16-BN, and ConvNeXt Tiny.

The role of each model is kept clear. EfficientNet-B0 is used as an efficient CNN

baseline. ResNet34 is used as a residual CNN baseline. MobileNetV3-Large is used as an efficient CNN classifier. DenseNet121 tests dense feature reuse. VGG16-BN acts as a classic CNN baseline. ConvNeXt Tiny represents a modern CNN design.

The final classification layer of each network is changed to five outputs. The models are trained with CrossEntropyLoss and AdamW. The learning rate is 5×10^{-4}. The weight decay is 10^{-4}. Cosine annealing is used. The batch size is 32. The maximum number of epochs is 15. Early stopping uses four epochs of patience.

The code stores the highest validation accuracy reached during training. These peak values are 81.31% for EfficientNet-B0, 80.49% for ResNet34, 83.49% for MobileNetV3-Large, 79.54% for DenseNet121, 81.17% for VGG16-BN, and 84.72% for ConvNeXt Tiny.

Based on these peak values, ConvNeXt Tiny and MobileNetV3-Large are selected for the ensemble. This selection value is not the same as the final report accuracy after the best-validation-loss weights are restored.

Table 3: Model Selection Based on Highest Observed Validation Accuracy

Model	**Peak Validation Accuracy**	**Final Report Accuracy**
EfficientNet-B0	81.31%	81%
ResNet34	80.49%	78%
MobileNetV3-Large	83.49%	83%
DenseNet121	79.54%	78%
VGG16-BN	81.17%	80%
ConvNeXt Tiny	84.72%	81%

3.9 Probability-Average Ensemble

The current notebook does not use a stack ensemble. It uses a two-model probability-average ensemble. This distinction is important for code and report alignment.

Let P_M be the MobileNetV3-Large probability vector. Let P_C be the ConvNeXt Tiny probability vector. Each vector has five class probabilities.

The ensemble probability vector is:

$$P_{\text{ensemble}} = \frac{P_M + P_C}{2}. \tag{1}$$

The final class is:

$$\hat{y} = \arg\max(P_{\text{ensemble}}). \tag{2}$$

Both models have equal weight. The notebook does not train a meta-learner. Therefore, this is not a stacking ensemble.

Summary

This chapter explained the full research method. It covered data loading, data partitioning, image processing, augmentation, normalization, class balancing, transfer learning, model selection, probability averaging, and Grad-CAM. It also explained why the ensemble is a probability-average ensemble and not a stack ensemble.

Chapter 4

4 Implementation and Result Analysis

This chapter presents the implementation and validation results of the proposed diabetic retinopathy classification framework. Six transfer-learning models are tested under the same broad training setup. Their results are compared with the two-model probability-average ensemble. Grad-CAM results are also discussed.

4.1 Experimental Environment

The notebook reports execution on CUDA. The implementation uses Python with PyTorch and TorchVision for model training. OpenCV is used for image reading and image processing. Albumentations is used for augmentation and normalization. Pandas reads the CSV labels. NumPy supports numerical operations. scikit-learn provides train-test split, classification reports, confusion matrices, and display functions.

The pipeline uses an NVIDIA GPU-enabled environment. The image size is 256×256. The batch size is 32. The maximum epoch count is 15. Early stopping patience is 4. AdamW is used with a learning rate of 0.0005 and weight decay of 0.0001. CrossEntropyLoss is used. The learning-rate schedule is cosine annealing.

4.2 Data Partitioning

The labeled APTOS data is split into training and validation subsets. The code uses an 80/20 split with stratification. The random state is fixed at 42.

The training part contains 80% of the available labeled data. The validation part contains 20%. Stratification is used because the five classes are not equally represented. It helps keep the class structure more consistent in both parts.

The validation report contains 733 images. The support values are 361 for Stage 0, 74 for Stage 1, 200 for Stage 2, 39 for Stage 3, and 59 for Stage 4.

There is no independent external test set in the current notebook. Therefore, all reported metrics in this chapter are validation results. They should not be described as external test results.

4.3 Evaluation Matrix

The final classification results are shown with accuracy, macro precision, macro recall, macro F1-score, and weighted F1-score. The figures below are kept as separate graphs.

Table 4: Comparison of Proposed Methodology with Existing Work

Author / Study	**Model Architecture**	**Accuracy (%)**
Pratt et al. [1]	Standard CNN	75.0
Wang et al. [2]	Inception-V3	63.23
Qummar et al. [7]	CNN Ensemble	80.8
Dondeti et al. [8]	NASNet + Non-linear Embedding	77.90
Proposed Work	**Ensemble (MobileNetV3-Large + ConvNeXt-Tiny)**	**83.0**

4.4 Accuracy Graph

The accuracy graph shows the final classification report values. MobileNetV3-Large and the ensemble both reach 83%. EfficientNet-B0 and ConvNeXt Tiny reach 81%. VGG16-BN reaches 80%. ResNet34 and DenseNet121 reach 78%.

The ensemble does not produce a higher final accuracy than MobileNetV3-Large in this run. Therefore, the result should be described as a matched accuracy result. It should not be described as an accuracy gain.

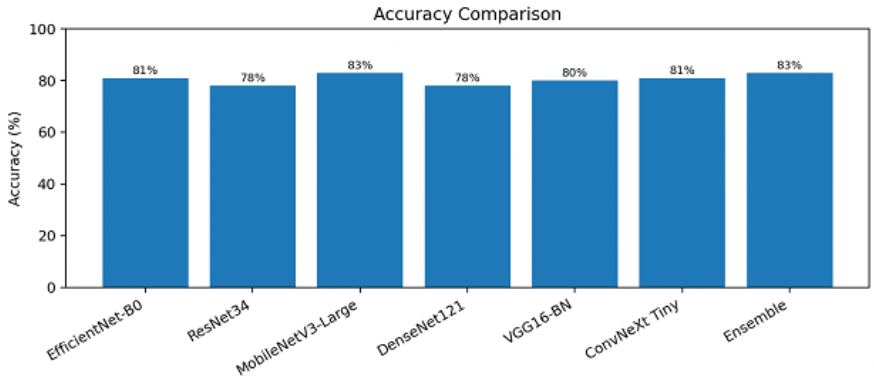


Figure 4.2: Accuracy comparison of the evaluated models.

4.5 Precision Graph

The macro precision graph shows 0.70 for MobileNetV3-Large and the ensemble. ConvNeXt Tiny reaches 0.69. ResNet34 reaches 0.68. EfficientNet-B0 and VGG16-BN reach 0.66. DenseNet121 reaches 0.64.

The values show that the probability-average ensemble keeps the same macro precision as MobileNetV3-Large in this run.

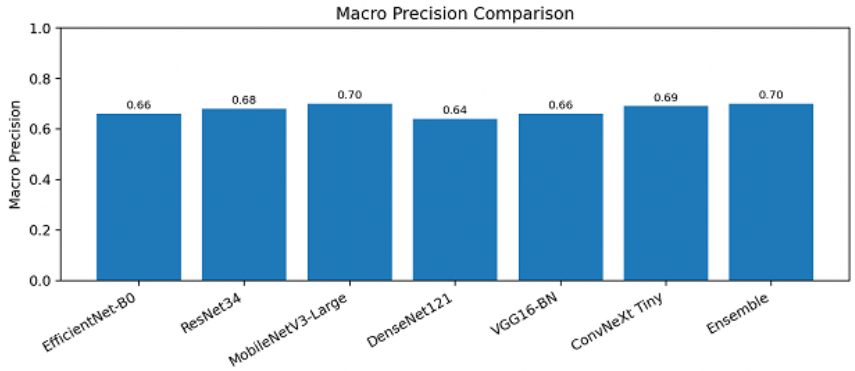


Figure 4.3: Macro precision comparison of the evaluated models.

4.6 Recall Graph

The macro recall graph shows 0.72 for MobileNetV3-Large, 0.71 for the ensemble, and 0.70 for ConvNeXt Tiny. VGG16-BN reaches 0.68. EfficientNet-B0, ResNet34, and DenseNet121 each reach 0.65.

The ensemble is close to the highest recall in the reported run. This shows that the combined output keeps a similar level of class finding ability.

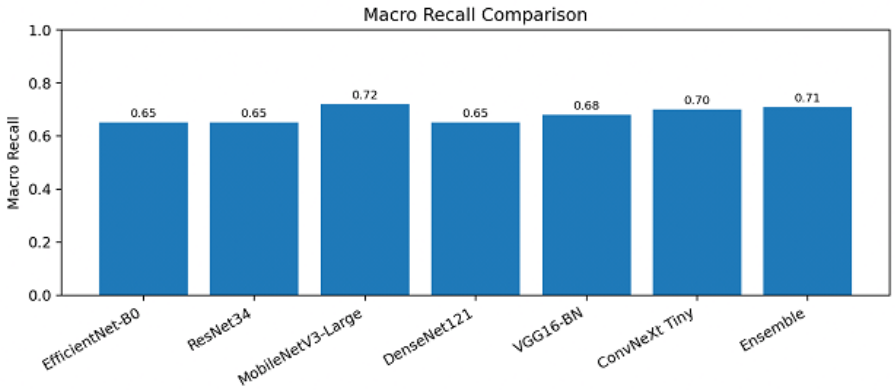


Figure 4.4: Macro recall comparison of the evaluated models.

4.7 F1-score Graph

The macro F1 graph shows 0.71 for MobileNetV3-Large and 0.70 for the ensemble. ConvNeXt Tiny reaches 0.68. VGG16-BN reaches 0.67. EfficientNet-B0 reaches 0.65. ResNet34 and DenseNet121 each reach 0.64.

The ensemble gives a result close to the strongest single-model value. This is useful when precision and recall must be viewed together.

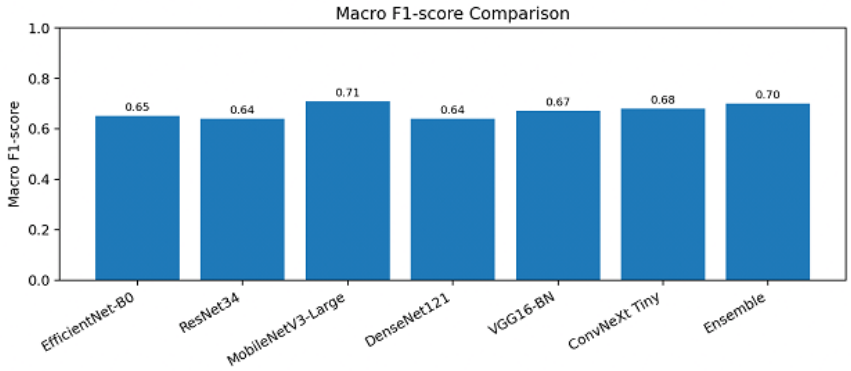


Figure 4.5: Macro F1-score comparison of the evaluated models.

4.8 Class-Level Results

The class-level reports show a clear difference between Stage 0 and the disease stages. Stage 0 has high precision, recall, and F1-score for the reported models. The smaller advanced stages are harder.

For MobileNetV3-Large, Stage 0 has F1 0.97. Stage 1 has F1 0.65. Stage 2 has F1 0.78. Stage 3 has F1 0.53. Stage 4 has F1 0.61.

For ConvNeXt Tiny, the F1 values are 0.97, 0.63, 0.74, 0.48, and 0.56 for Stages 0 to 4. For the ensemble, the values are 0.97, 0.64, 0.77, 0.54, and 0.56.

Stage 3 has the lowest support with 39 validation images. It also has low F1 values. This shows why macro metrics are useful. They give equal weight to all five classes.

Stage 1 recall is relatively high for MobileNetV3-Large at 0.80 and for the ensemble at 0.77. This suggests that the models can find many Stage 1 samples. Precision remains lower because some samples from other stages are predicted as Stage 1.

4.9 Confusion Matrix

The ensemble confusion matrix is generated directly by the notebook. It compares the true five stages with the predicted five stages.

The diagonal cells represent correct predictions. The off-diagonal cells represent class confusion. The classification reports show that the model is strongest on Stage 0 and Stage 2. Smaller disease stages have more errors.

4.10 Training Behavior

Early stopping occurred at different epochs. EfficientNet-B0 stopped after 7 epochs. ResNet34 stopped after 9 epochs. MobileNetV3-Large stopped after 10 epochs. DenseNet121 stopped after 6 epochs. VGG16-BN stopped after 11 epochs. ConvNeXt Tiny stopped after 11 epochs.The code restores the weights linked to the lowest validation loss before producing the final classification report.This explains why final report accuracy can differ from the maximum validation accuracy seen during training. For example, ConvNeXt Tiny reaches 84.72% at one validation epoch. Its restored lowest-loss model gives 81% in the final report.The code therefore uses two values for two purposes. Peak validation accuracy is used for ensemble selection. Lowest-validation-loss weights are used for the final report.

4.11 Ensemble Result

The ensemble uses ConvNeXt Tiny and MobileNetV3-Large. Its final report gives 83% accuracy, macro precision 0.70, macro recall 0.71, and macro F1-score 0.70. Its weighted precision, recall, and F1 are all 0.83.The ensemble should not be described as an accuracy improvement over MobileNetV3-Large. Both have 83% final accuracy. The contribution of the ensemble is the combination of two probability outputs under one decision rule.

4.12 Grad-CAM Results

The notebook generates one original image and one Grad-CAM activation image for each of the five classes. The explanation is produced from ConvNeXt Tiny. The

heatmap shows regions that contribute to the chosen class score. The Grad-CAM output is a visual inspection tool for the learned representation. It should not be interpreted as a clinical lesion detector.The code does not compare the heatmap with expert lesion masks. Therefore, the heatmap can show model focus. It cannot by itself prove that the model has identified a medically correct lesion.

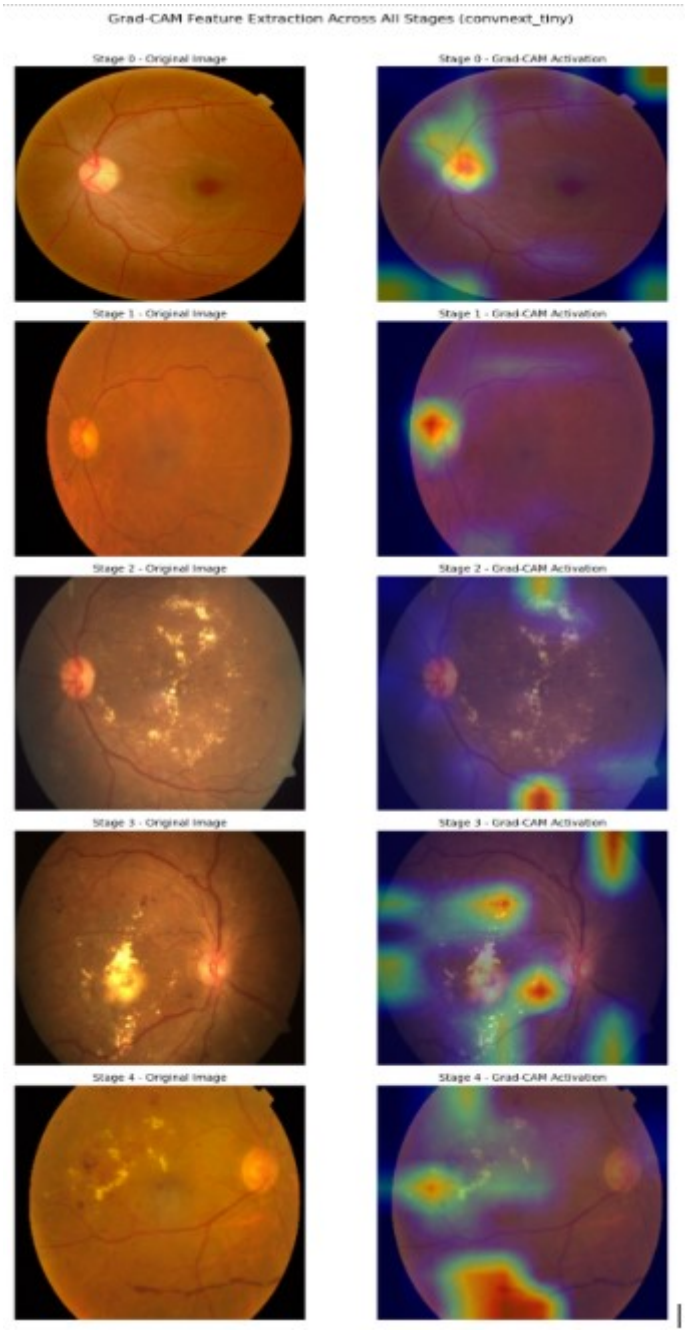


Figure 4.6: Grad-CAM examples for the five DR stages.

4.13 Result Discussion

The results show that all six transfer-learning models can learn the five-class task under the same broad training setup. Their final accuracies range from 78% to 83%. The comparison is useful because all models use the same validation set and similar training controls.MobileNetV3-Large gives 83% final accuracy. It also gives macro precision of 0.70, macro recall of 0.72, and macro F1 of 0.71 in the reported classification results. ConvNeXt Tiny has a different result. It reaches the highest observed validation accuracy during training at 84.72%. The code selects it for the ensemble. Its restored final report accuracy is 81%.The ensemble reaches 83% accuracy. It therefore matches MobileNetV3-Large. It does not exceed it. Its macro recall of 0.71 and macro F1 of 0.70 are close to the single-model values.The class-level results also show that the five-class task is difficult. Stage 0 is easier than the smaller disease classes. Stage 3 has only 39 validation samples. It also has low F1 values. This supports the need for more balanced data and stronger class-aware methods in future work.

4.14 Comparison With Earlier Work

The present results should not be compared as if they were the same task as all earlier studies. Gulshan et al. focused on referable DR detection and used EyePACS-1 and Messidor-2. Lam et al. focused on lesion detection from image patches. Pratt et al. studied CNN-based DR grading.The present work uses the APTOS 2019 dataset, five-class grading, six transfer-learning models, a two-model probability-average ensemble, and Grad-CAM.Therefore, the main comparison is methodological. It is not a direct ranking of accuracy. The data, label definitions, validation design, and task targets differ across studies.

Summary

This chapter presented the implementation and validation results. It described the experimental environment and the data partition. It then presented the evaluation

matrix and four separate metric graphs. Class-level results, confusion behavior, training behavior, ensemble results, and Grad-CAM results were also discussed. The chapter keeps peak validation accuracy separate from final checkpoint accuracy.

Chapter 5

5 Standards, Constraints, Ethics, and Challenges

The development of a medical artificial intelligence system needs clear engineering practice. It also needs careful use of data and results. This chapter discusses the software standards, social impact, ethics, constraints, and main challenges of the proposed framework.

5.1 Software and Engineering Standards

The implementation uses established open-source libraries. PyTorch provides the deep learning framework. TorchVision provides pretrained model weights. OpenCV handles image input and enhancement. Albumentations handles image transformations. Pandas manages label data. NumPy supports numerical operations. scikit-learn provides the data split and evaluation functions.The use of a fixed random state of 42 for the stratified split supports repeatability. The common batch size, loss function, optimizer family, learning-rate schedule, and early-stopping rule also make the six-model comparison more controlled.The code saves the best model weights during training. It restores the weights linked to the lowest validation loss before the final classification report. This keeps the evaluation process clear and repeatable.

5.2 Constraints

The project has several limits.

- Only one public dataset is used.
- The notebook uses one 80/20 training-validation split.
- There is no separate external test set.
- Images are reduced to 256×256 pixels.

- Some disease classes have much lower support than Stage 0.
- Six deep models require substantial GPU memory and training time.
- The ensemble uses equal probability weights rather than learned weights.
- Grad-CAM is qualitative and is not checked against expert lesion masks.

The design also has a fixed five-class output. This follows the APTOS task. The model input is fixed at 256×256 pixels so that all six CNNs can use the same broad input setting.

5.3 Ethical Considerations

Medical image classification requires careful use. The model is a research system. It should not be used as an independent medical decision maker.A real clinical system would need external validation, expert review, safety testing, calibration, privacy controls, and regulatory assessment. An incorrect disease stage may affect a screening decision. The model output should therefore be treated with care.The study uses the public APTOS 2019 Blindness Detection dataset under its applicable research terms. The project does not add personal patient information. Even so, research results should be handled with appropriate care.

5.4 Social Impact

Automated retinal image analysis may support screening workflows where many images need review. A model can provide a first classification and a visual explanation. This may help a trained reviewer organize cases for further review.

The value of such a system depends on its performance on new populations and imaging devices. A model trained on one dataset may not behave in the same way on every new setting.

The system should therefore be viewed as decision support. It should not replace an ophthalmologist or another qualified health professional.

5.5 Main Challenges

Several challenges were found during the work.

First, nearby DR stages can look similar. This can lead to class confusion. Second, some advanced stages have few validation samples. Stage 3 has only 39 validation images in the reported split. Third, retinal images can vary in light, focus, color, and image quality.

Another challenge is the difference between peak validation accuracy and final checkpoint accuracy. ConvNeXt Tiny reaches 84.72% at one validation epoch. Its restored final report gives 81%. The code uses these two values for different purposes.

Training six pretrained models also needs high GPU cost. Another challenge is the choice of a suitable Grad-CAM target layer for each model. These issues affect both system design and result analysis.

5.6 Project and Resource Constraints

The project was designed around available software and computing resources. The main software tools are Python, PyTorch, TorchVision, OpenCV, Albumentations, Pandas, NumPy, and scikit-learn.

The use of a public dataset and open-source software helps reduce project cost. The training work still needs an NVIDIA GPU-enabled environment because six CNN models are trained.

5.7 Project Timeline

The capstone research project was executed over a six-month period from November 2025 to April 2026. The first phase focused on literature review, data acquisition, and environment setup. The next phase focused on image processing and initial model tests. Later phases covered model training, ensemble development, Grad-CAM, result analysis, and report writing.

Table 5: Project Timeline and Milestones

Phase / Month	**Key Activities**
November 2025	Literature review, dataset acquisition, environment setup
December 2025	Image processing, data augmentation, initial model testing
January 2026	Training EfficientNet, ResNet, DenseNet, VGG16 models
February 2026	Training MobileNetV3 and ConvNeXt, hyperparameter tuning
March 2026	Ensemble development and Grad-CAM integration
April 2026	Final result analysis, figures, and report writing

Summary

This chapter discussed the engineering and social context of the framework. It covered software standards, constraints, ethics, social impact, challenges, resources, and project timing. The main limits are the single dataset, one validation split, class imbalance, fixed image size, equal ensemble weights, and qualitative Grad-CAM analysis.

Chapter 6

6 Conclusion

This research developed an explainable transfer-learning framework for five-class diabetic retinopathy grading. The implementation uses the APTOS 2019 Blindness Detection dataset. The code uses an 80/20 stratified split, 256×256 image processing, Gaussian image enhancement, training-time geometric augmentation, ImageNet normalization, and weighted random sampling.Six pretrained CNN models were trained. They are EfficientNet-B0, ResNet34, MobileNetV3-Large, DenseNet121, VGG16-BN, and ConvNeXt Tiny. The final classification reports give 81%, 78%, 83%, 78%, 80%, and 81% accuracy, respectively.MobileNetV3-Large gives 83% final accuracy. ConvNeXt Tiny reaches the highest observed validation accuracy during training at 84.72%. The code selects ConvNeXt Tiny and MobileNetV3-Large for the ensemble because they have the two highest observed validation accuracy values.The ensemble averages the two softmax probability vectors. It gives 83% accuracy, macro precision 0.70, macro recall 0.71, and macro F1-score 0.70. It does not improve final accuracy over MobileNetV3-Large in this run. Its contribution is the combination of two model outputs under one decision rule.Grad-CAM is applied to ConvNeXt Tiny. It produces heatmaps for the five DR stages. These heatmaps provide a visual view of model focus. They do not prove clinical correctness because the study does not compare them with expert lesion masks.Overall, the code demonstrates a complete research pipeline from data loading to explainable prediction. It also shows clear areas for further work. The lower scores on smaller disease classes show that stronger class-aware learning and broader validation are still needed.

6.1 Limitations

The evaluation is based on one validation split from one public dataset. There is no independent external test set. The five disease classes are also imbalanced. The

smallest validation class has only 39 images.

The final image resolution is limited to 256×256 pixels. The ensemble uses only two models and equal probability weights. The final reports use restored lowest-validation-loss weights, while ensemble selection uses peak validation accuracy.

Grad-CAM is qualitative. It is not clinically validated in this study. The heatmaps are therefore useful for model inspection, but they should not be treated as clinical lesion masks.

6.2 Future Work

Future work should address the current limits in a clear way.

- Use multiple external datasets for cross-dataset validation.
- Use a separate held-out test set that is never used for model selection.
- Study class-aware loss functions and stronger sampling methods.
- Test stronger image quality control and lesion-aware preprocessing.
- Tune ensemble weights instead of using equal averaging.
- Test larger or more specialized retinal models.
- Evaluate calibration and confidence estimates.
- Compare Grad-CAM maps with expert lesion annotations when available.
- Study model compression for lower-cost deployment.
- Perform clinical validation before any real-world use.

Future work can also study how the framework behaves on new populations and imaging devices. This can help test whether the learned features remain useful outside the current dataset.

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ing, vol. 15, no. 8, pp. 1679–1686, 2021, reported five-class validation accuracy: 75.50%.

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ANNEXURE–I

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Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (it may misidentify writing that is likely human generated as AI generated and likely AI generated as human generated) so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

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